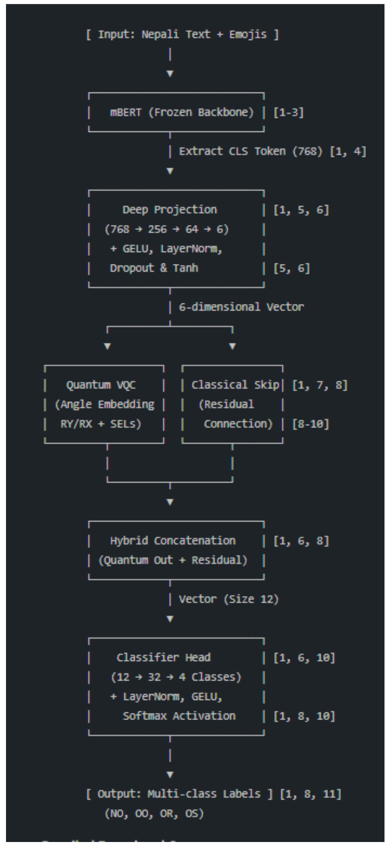


1. **Feature Extraction (Classical mBERT)**

- Input : The model accepts Nepali text, preserving emojis.
- Backbone : It uses mBERT which is kept frozen to act as feature extractor.
- Output : The model extracts the CLS tokens embedding, a 168-dimensional vector representing the entire sequence.

2. **Deep Classical Projection**

- Before reaching the quantum layer, the 768-dimensional embedding undergoes a “Deep staged projection” to gradually compress the data while preserving information.

  Layer 1: Linear (768 → 256), LayerNorm, GELU activation, and Dropout (0.3) .
  Layer 2: Linear (256 → 64), LayerNorm, GELU activation, and Dropout (0.3) .
  Layer 3: Linear (64 → n_qubits), followed by a Tanh activation to squash values into the [−1,1] range required for quantum embedding

3. **Quantum Variational Layer (VQC)**

   The compressed 6 dimensional vector is processed by quantum circuit designed for high expressivity.

   - Encoding : Features are encoded into quantum state using AngleEmbedding on both the **RY** and **RX** rotation axis.
   - Variational Layers : The model uses StronglyEntanglingLayers (3 layers across 6 qubits) to create entanglement and apply variational parameters (54 total quantum parameters)
   - Measurement : All 6 qubits are measured in the PauliZ basis, returning expectation values that serves as quantum output.

4. **Residual and Hybrid Fusion**
   The architecture includes a classical skip connection (residual path) to ensure stable gradient flow and prevent information loss from the quantum layer.

   - **Residual Path:** A linear projection (Linear(64, 6)) runs parallel to the VQC.
   - **Concatenation :** The quantum output is normalized and then concatenated with the residual output, resulting in hybrid feature vector (size 12)

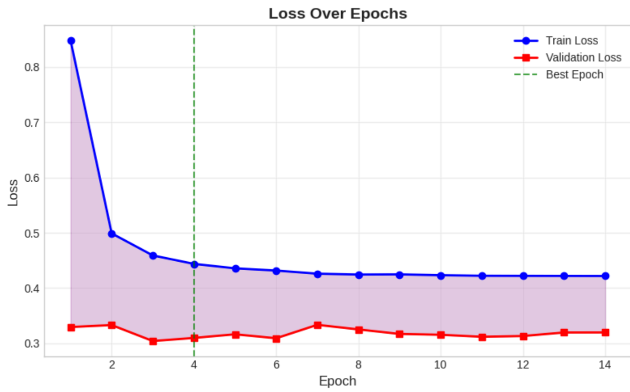
5. **Final Classifier Head**
   The fused hybrid vector is fed into a wider classifier head to produce final predictions:
   - Layers : Linear (12- 32), LayerNorm, GELU, Dropout, and a final Linear (32-> N classes)
   - Output : A Softmax function provides possibilities for each classes (eg.,NO,OO,OR,OS)

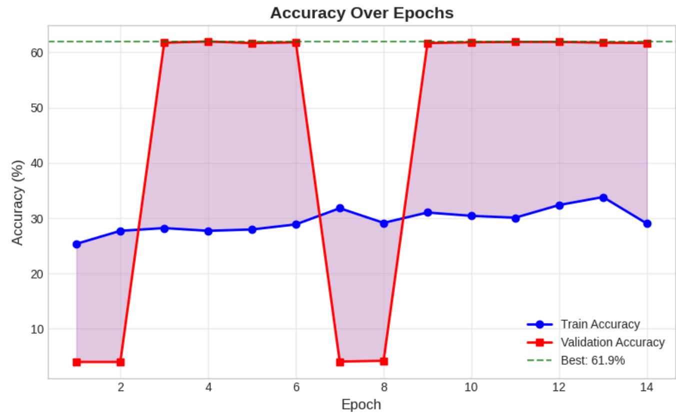
Outputs

When computing the model the following results were obtained

Category	Precision	Recall	F1-Score
NO (Non-hate)	0.6267	0.9587	0.7580
OO (Offensive)	0.4938	0.0821	0.1408
OR (Religious)	0.00	0.00	0.00
OS (Sexual)	0.00	0.00	0.00
Overall Accuracy	-	-	0.6193

Graphs :





Conclusion :

Though the result is not satisfactory by any means, this is simply a concept testing of the hybrid mBERT and VQC model. This model can be enhanced further

- **Fine-tuning the quantum VQC part :** This includes exploring different encoding schemes beyond the current **RY** and **RX** AngleEmbedding or increasing the number of StronglyEntanglingLayers and qubits beyond the current 3-layer, 6-qubit configuration to increase expressivity
- **Tuning the mBERT parameters** : The current architecture uses a frozen mBERT backbone purely for feature extraction . Unfreezing these layers for full fine-tuning on the Nepali dataset could allow the model to learn more task-specific Devanagari and emoji nuances.
- **Optimizing the feature projection bottleneck :** The current deep staged projection compresses a 768-dimensional embedding down to 6 dimensions to fit the quantum circuit . Adjusting this compression ratio or the specific architecture of the linear layers (768 → 256 → 64 → 6) could help preserve more information before it reaches the quantum layer .